CATERPILLAR SUPPLIER QUALITY MANUAL

VERSION 3.0

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Caterpillar: Confidential Green

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Interactive Functionality

For the functionality to work correctly, please view in Adobe Acrobat Reader.

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A MESSAGE FROM PAM

At Caterpillar, we strive to ensure every product, service and solution exceeds the expectations of our customers. This strategy focuses on delivering Sustainable Growth governed by our Values in Action and the Operating and Execution model. As a key supplier, we ask that you leverage your component expertise and drive continuous improvement to ensure zero defects in everything we do.

This manual defines our supplier expectations and incorporates information and links to support a zero-defect methodology across our portfolio of brands and class leading products. Working in partnership, we will continue to mutually grow our businesses and deliver exceptional products to our customers.

Pamela Heminger
Vice President, Strategic Procurement
& Planning Division (SPPD)

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DOCUMENT PURPOSE (ROLL OVER TO VIEW)

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INTRODUCTION

Caterpillar is the world's leading manufacturer of construction and mining equipment, diesel and natural gas engines, industrial gas turbines and diesel-electric locomotives. We are a leader and proudly have the largest global presence in the industries we serve.

At Caterpillar, we take pride in what we do and what we make possible – from the quality of our products and services to the people who stand behind them. But what matters most to us is how we help customers build a better world.

Our strategy defines the next chapter in Caterpillar's legacy, and we continue to build on the foundational principles of Our Values in Action and the Operating and Execution Model.

Our people have always been the core of Caterpillar. Together with our partners, we continue to develop lifelong relationships with our customers.
The quality of our products and services reflects the power and heritage of Caterpillar, the pride we take in what we do and what we make possible.
We are passionate about people, process, product, and service excellence. We are determined to serve our customers through innovation, continuous improvement, an intense focus on customer needs and a dedication to meet those needs with a sense of urgency.

Caterpillar suppliers are key partners in developing solutions for our customers, from product development through each component's life cycle. Together, we help make our customers more successful by developing lifelong relationships and earning trust and loyalty through a superior customer experience.

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SOURCING AND ONBOARDING

Selecting suppliers for Caterpillar is one of the most important decisions we make. Caterpillar suppliers have a direct impact on our customers, competitiveness, brand, and ability to deliver on future developments. We value the relationships we build with our suppliers and from the first step, we want to be sure of a solid foundation.

QUALITY SYSTEM REQUIREMENTS

All suppliers are expected to have a Quality Management System (QMS) documented, assessed, and certified as compliant against ISO 9001:2015 as a minimum with IATF 16949:2016 seen as an alternative where applicable. In addition to the QMS certification, it is expected that suppliers demonstrate their commitment to the environment with an Environmental Management System that meets the requirements of ISO 14001 or equivalent.

SAFETY

Our safety culture extends beyond our internal operations to include the health and safety of everyone in, on or around our products. Our commitment to safety includes the design and engineering of our products and extends to operator training, certification, and solutions to improve workplace safety culture.

For Caterpillar employees visiting suppliers' facilities or for suppliers visiting a Caterpillar facility, it is expected that all representatives follow the required safety protocols.

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GETTING STARTED

In accordance with our Quality Management System and to ensure we select only the best suppliers for our products, facility and business audits will be completed using the Quality, Cost, Logistics, Development and Management (QCLDM) framework. This audit will confirm correct processes, procedures, and stability within your business to support our needs as a chosen supplier. This QCLDM baseline will then form an integral part of our ongoing Supplier Performance Management (SPM) process.

If required, additional technical, process or product audits will be carried out depending on the commodity or complexity of product being supplied. It is important that adequate resources are in place during these audits to ensure timely completion and effective assessment of capabilities.

To access Caterpillar supplier facing portals and systems, a Corporate Web Security (CWS) login is required for nominated representatives. Access will be granted by an Identity Manager (IDM) and delegate, who will be identified by the supplier, with training available through Supplier Development College (SDC). Once CWS access has been granted, Supplier Connect can be accessed with links to tools and information required to do business with Caterpillar. The tab titled "**Doing Business with Us**" defines our core expectations for engaging with our suppliers.

TRAINING

Quality is a core value at Caterpillar, and we want the same from our suppliers. You will be expected to have trained and competent employees at all levels within your business to ensure you deliver the best quality and have the right capabilities to collaborate during all aspects of doing business.

Training opportunities are available for our suppliers as we build on our strategic relationships and align processes with best practices developed by Caterpillar. Your procurement representatives will be able to guide you if required, however Supplier Development College provides some useful training material to guide you through our core processes.

IDENTITY MANAGER REQUESTS
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SUPPLIER DEVELOPMENT COLLEGE
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SUPPLIER CODE OF CONDUCT

At Caterpillar, our internal Code of Conduct is the foundation of the work we do. It defines our Values in Action – what we stand for and how we conduct ourselves with customers, suppliers, and one another. Whilst we conduct business within applicable laws and regulations, our values set clear expectations beyond the law for ourselves and business partners.

Caterpillar suppliers are an extension to the value chain and our commitment to quality. We ask all suppliers to understand our Code of Conduct, follow applicable laws, uphold the sound business practices Caterpillar embraces, and to conduct business in a manner that respects human rights.

REGULATORY COMPLIANCE

Caterpillar operates globally and conducts business within applicable laws and regulations. At times, Caterpillar may request suppliers to provide information reflecting compliance with those laws and regulations, and responding to those requests is a requirement for doing business with us.

SUPPLIER CODE OF CONDUCT
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CONFLICT MINERALS
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CHEMICAL REGULATIONS
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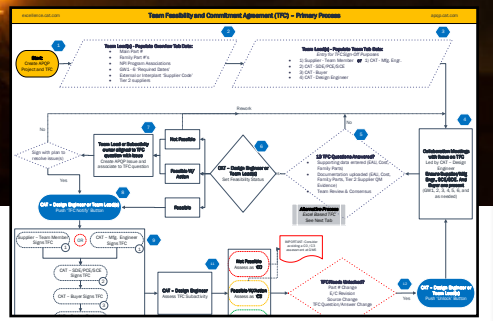


ROLL OVER FOR TFC PROCESS FLOW

PROCESS ADHERENCE

Caterpillar understands the value of robust processes and the opportunity to drive predictable excellence within our value chains. Caterpillar suppliers will be expected to have controlled processes and the trained resources available to support, lead and engage with us at every step of the way.

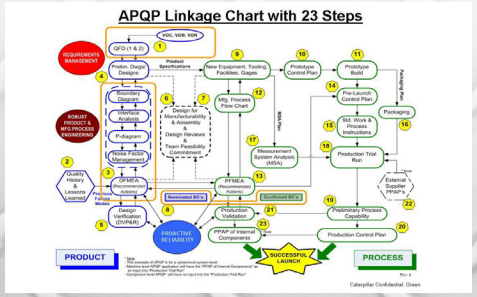
ADVANCED PRODUCT QUALITY PLANNING (APQP)

Advanced Product Quality Planning (APQP) has been deployed using the principles outlined in the AIAG APQP Manual. It is expected that resources trained and experienced in APQP, will be allocated to lead projects aligned with Caterpillar New Product Introduction (NPI) programs and current production sourcing activity.

It is expected to utilise an APQP process aligned to the AIAG manual for all components and projects aligned with an NPI Program, PCN, Supplier Resource or Fit, Form or Function Design Change. It is also expected for suppliers to utilise the same robust APQP methodology with their suppliers to ensure quality requirements are followed throughout the value chain.

Where directed or requested by Caterpillar, APQP projects will be completed within the Caterpillar APQP management tool (EtQ Reliance). Caterpillar may provide or make available training where appropriate. Milestones for project completion will be defined by your Caterpillar representative, i.e., Buyer, Supplier Development Engineer, etc., and it is expected to maintain conformance of activities to these timelines.

ROLL OVER FOR APQP FLOW CHART



To ensure accountability and to drive collaboration, the Team Feasibility and Commitment (TFC) agreement will be used to ensure the project delivers a product that is feasible to manufacture, at the right cost, within specifications, at capacity and on time. Sign off from the APQP team (Caterpillar and Supplier) is required at Gateway 3, however the contents of the document and agreement should be referenced through all project gateways to maintain progress on key actions.

APQP EXPECTATIONS
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APQP MANAGEMENT TOOL
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PRODUCTION PART APPROVAL PROCESS (PPAP)

Caterpillar and its Quality Management System are aligned with the AIAG PPAP manual. At a facility level, all supplied production components are expected to have an approved PPAP within the Caterpillar PPAP system (EtQ Reliance). Specific documentation requirements will be outlined by assigned Caterpillar representatives; however, it is expected that all required documents will be stored within a supplier's own PPAP system for review on demand.

A PPAP will be required for any of the following situations:

- New Component
- Engineering Changes (Supplier or Caterpillar led dependent on design control)
- Tooling: Transfer, replacement, refurbishment or additional
- Correction of a discrepancy from a previous submission
- Inactive tooling (> 1 year)
- Change to optional construction or material
- Sub Supplier or Material source change
- Change in component processing method or supplier
- Change of manufacturing location or additional location

A PPAP request will be generated by Caterpillar with the level of submission defined through a risk assessment of the change (as above list). For supplier led process changes submitted via a Process Change Notification (PCN), the supplier should initiate the PPAP with the level of submission being defined via approval of the PCN.

PPAP EXPECTATIONS
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ROLL OVER FOR LAYERED AUDIT INFO

PRODUCTION PART APPROVAL PROCESS (PPAP) (CONTINUED)

PPAP documentation will be submitted by the supplier to evidence the processes and materials used to manufacture the product in accordance with the design records (Engineering Drawing) and related technical specifications (Caterpillar 1E specifications, International Standards etc). The processes and materials shall be those intended for full volume production, capable of delivering the required run at rates. Approval of PPAP documents will not be made in full if prototype tooling, off production processes or other development activities have been used in manufacture, however interim approval may be used in these circumstances to allow validation and project steps to be completed. Components should not be shipped to a Caterpillar facility unless at least an interim approval has been given to a submitted PPAP.

Whilst the application of PPAP is to document the processes and resulting conformance of product to meet production capability, it may on occasions be requested that PPAP documents are submitted to support the steps of a project. If this level of submission is required, suppliers will be requested to support submissions of phased or pre-production documentation.

PROCESS CONTROL EXPECTATIONS

Successful application of APQP and PPAP has an output of well-defined processes and controls. Caterpillar has deployed these core supplier quality processes as they limit variation, prevent defects, and provide a foundation for audits. We expect a never-ending pursuit of excellence in everything we do, and process control and capability support this strategy..

To maintain confidence in process, control a layered auditing process should be considered. Layered Process Audits (LPA) can be performed at any level of the business; however, Caterpillar encourages suppliers to engage through all levels of the organisational structure to demonstrate management commitment.

For certain materials, processes, or component types Caterpillar may require further ongoing product and process audits. These commodity specific audits will typically be performed at the site of manufacture with adequate resources made available to complete in a timely manner. These audits are used to periodically confirm that all required technical specifications continue to be met and that manufacturing processes are consistent with those held in the latest approved PPAP.

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PROCESS CAPABILITY

Special characteristics can be defined as process or product parameters, features or characteristics that relate directly to regulatory compliance, a critical customer requirement, critical business requirement, chronic product issue, or has a parameter that needs to be closely controlled to ensure that a component or assembly complies with specifications that effect a special design or special required characteristic.

Caterpillar defines special characteristics on the design record for each component or assembly in accordance with 1E2966. Where identified during APQP, Supplier Resourcing or an Engineering change, it is expected that adequate controls are defined and documented within the control plan.

1. Special Required Characteristic = Product or Process parameter that directly affects compliance to a governmental or regulatory requirement
2. Special Design Characteristic = Product characteristic where variation could significantly impact the products safety or customer satisfaction
3. Special Process Characteristic (Design) = Process parameter identified within the DFMEA where a specific process must be used or where the characteristic will be difficult, impractical, or impossible to measure directly
4. Special Process Characteristic (Supplier) = Product or process characteristics identified by the supplier where the impact of variation could impact a later process or assembly–These characteristics are dynamic and will be documented within the PFMEA and Control Plan. Any features identified as supplier nominated special characteristics will be managed by the suppliers manufacturing and engineering teams.

In pursuit of excellence, Caterpillar expects identified Special Characteristics to meet a minimum 1.33 Cpk, however the target should be at least 1.67 Cpk. Where features cannot be measured to give a Cpk (one sided tolerance, go/no go, etc.) approval for the equivalent control should be determined with the relevant Design Control and SDE. All controls and defined features should be documented in the supplier control plan and PFMEA accordingly. If a Cpk of at least 1.33 cannot be achieved, include increased inspection (typically 100%) agreed by both parties during PPAP submission.

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PROCESS CHANGE NOTIFICATION (PCN)

Caterpillar acknowledges that in a dynamic working environment there may be occasions whether planned or unplanned where processes, materials, suppliers, etc., will change. Caterpillar utilises a Process Change Notification (PCN) to document these changes and either approve or reject the requested changes. As a supplier you will be required to give at least 6 weeks' notice for any change from the list below (not limited to) or when requested by Caterpillar representatives. Emergency PCN submissions will be considered depending on the risk identified because of the change. It may be necessary for such requests to be escalated for review.

- A planned process change, e.g., tooling change, machine change, process flow change.
- Emergency and temporary changes to an approved manufacturing process.
- Manufacturing changes identified through sub tier suppliers that affect Tier 1 suppliers.
- A minor change in the material that is already authorized within the range specified by a 1E specification.
- A skilled labour change at the supply base which may affect an approved manufacturing process.

It is the responsibility of the supplier to ensure that any validation requirements outlined by Caterpillar resulting from the submitted PCN are met prior to future supply. For the majority of PCN requests, an updated PPAP will also need to be submitted to document the changes made in the manufacture of affected products. The specific requirements will be agreed as part of the validation plan and should be uploaded to a revised PPAP submission (in EtQ Reliance) with approval granted by Caterpillar prior to shipping product subject to the change.

To validate and assure the effectiveness of suppliers process control, Caterpillar will, on occasion, perform PCN audits. It is expected that all current approved material suppliers, manufacturing processes and supporting documentation being used at the time of audit is in accordance with the latest approved PPAP held by Caterpillar. Any non-conformance found will require a corrective action plan, and if requested, further supplier led audits to confirm that any other Caterpillar product manufactured is in accordance with the latest approved PPAP.

PCN RESOURCES
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ROLL OVER FOR ADDITIONAL PCN CRITERIA

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CATERPILLAR BRANDING REQUIREMENTS

Caterpillar is proud of the quality that goes into every part used in our products, and it is appropriate in most cases to identify these parts using one of the Caterpillar product brand trademarks or word marks. Detailed information related to Caterpillar branding requirements for production parts and/or aftermarket components can be found on the respective engineering drawing and tied to our engineering specification **1EO198**. The use of Caterpillar branding for any other purpose must meet the requirements of our defined branding criteria with further information available at **global brands**.

TOOLING

An integral component of any robust and capable process is quality tools. Caterpillar will purchase and retain ownership of tooling, if used exclusively by suppliers for the manufacture and quality assurance of our products, and typically covers the initial tools and any subsequent tools required to support the product life cycle. Costs may be shared to protect Supplier Intellectual Property, or in the case of certain forging applications and Steel Mill Industries, the tooling will be retained by the manufacturer with costs expensed to Caterpillar where applicable.

To manage its tooling processes Caterpillar has developed a Supplier Tooling Management System (STMS) where all tooling transactions can be completed. From a supplier perspective, the STMS is where you can add quotes for requested tooling, update tool life expectations and condition (1-5 scale), provide images of tooling and confirm asset tag allocation. The Caterpillar buyer will use the inputs within STMS to monitor new and refurbishment tooling requirements to proactively manage tooling, mitigating any potential assurance of supply issues.

Maintaining records of tooling including suppliers, costs, life expectancy and condition is vitally important to ensure manufacturing processes deliver quality products. It is expected that suppliers will have defined processes to continuously monitor tool condition and provide Caterpillar with the required details as outlined above

COMPONENT BRANDING
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STMS SUPPLIER CONNECT
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STMS RESOURCES
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SUB SUPPLIER MANAGEMENT

Within this manual, Caterpillar defines the expectations for Supplier Quality from suppliers. In order to assure quality through the supply chain, we have an expectation that the same requirements will be cascaded through the tiers of supply. As described throughout this manual, our tools, processes, procedures all work together to build the strategic relationships required to deliver quality products and this should be no different to the relationships you have with suppliers. During onboarding and through ongoing performance management, we will look for processes used to actively mitigate risk through the supply chain.

PACKAGING

The packaging of product destined for Caterpillar facilities (Production Material) or our customers and dealers (Aftermarket Parts) should be capable of preserving the quality level of the product from manufacture to point of use. Packaging materials should be capable of preventing damage, corrosion, and contamination during transit and storage.

During product and process development, packaging requirements should be documented as part of the PPAP submission. Packaging specifications need to take into consideration the individual requirements of receiving facilities or aftermarket customers and will often lead to unique specifications depending upon the customer and intended supply chain. The chosen packaging specifications form part of the approved manufacturing process and as such any changes need to be approved by each customer prior to implementing.

Further information related to Caterpillar packaging requirements can be found within our Packaging manual here.

PACKAGE & SHIPPING MANUAL
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NONCONFORMANCE MANAGEMENT

Caterpillar has deployed Lean methodology across our businesses and expects that suppliers will join us on this journey. The Caterpillar lean model places Built in Quality (BiQ) as the first pivotal step in the process and emphasises the value defect-free manufacturing provides the business and our customers.

As a customer, our expectations cascade through the BiQ model to focus on stopping defects before reaching the next customer. We have deployed quality gates, Pre-Delivery Inspection (PDI), Inbound Quality Audit (IQA), In Process Validation (IPV), and Pre-Delivery Audit (PDA) intentionally within our businesses to capture defects and pass them back to the area where they were created. These key activities are core to a process of ongoing gap analysis and improvement covered under Customer Acceptance Validation (CAV). The intent of CAV tools is to ensure measurement, monitoring and key manufacturing processes are continuously improved to meet the needs of our customers. CAV Assessments may be completed on supplier processes to further eliminate the risk of defective material reaching Caterpillar facilities.

SUPPLIER ACCOUNTABILITY

Caterpillar processes are focussed on delivering defect-free products and we expect the same commitment from our suppliers and sub suppliers. You should provision sufficient qualified resources to support the manufacturing process, and customer to deliver quality products. If nonconforming material is identified prior to despatch, robust processes for managing the risk of shipping defective product, and implementing corrective actions should be used. Where defective product is found at a Caterpillar facility or customer, we expect the supplier of material to lead investigation, containment and corrective action. Further details are contained in subsequent sections of this manual.

As a result of poor quality, we also expect suppliers to be accountable for managing any additional impacts that may result from the identification of defective product. Whilst not exhaustive this can include additional stock, increased inspection, rework, emergency freight, actions from sub suppliers etc.

NCM SUPPLIER CONNECT
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CORRECTIVE ACTION

If nonconforming material is identified within a Caterpillar facility, the following core expectations are required to be performed by our suppliers:

- Emergency Containment Action (ECA) – Containment - is expected within 24 hours or earlier depending upon the severity of the nonconformance or impact and risk to our operations, product, or customer. The initial communication of a nonconformance will be communicated by the facility Supplier Quality Engineer (SQE) or equivalent. If unable to support within initial containment actions a third-party inspection provider can be used, or Cost of Poor Quality (COPQ) will be captured if Caterpillar labour is required

- Interim Containment Action (ICA) – STOP Gate - is required to be completed once an ECA has been initiated and will usually be in the 24-48 hours following initial defect communication. Aligned with our BiQ principles, the priority is to stop the flow of further defects and implement a gate or inspection point to prevent further flow of defects. This gate can be at the supplier's discretion, however following a nonconformance it is normal for increased Inbound Quality Audit (IQA) frequencies to be implemented within Caterpillar and repeat defects will be identified as poor implementation of stop gates. Your Caterpillar supplier quality contact may, at this point, instruct that a third party be installed or your own representative to 100% inspect product prior to release at the Caterpillar receiving facility.

- Permanent Corrective Action (PCA) – Resolution – is the identification of the root cause/causes of a nonconformance and the action plan for process correction. Caterpillar expects suppliers to follow a documented 8D process or equivalent and maintain records for all reported nonconformances. A corrective action plan is required within a minimum of 14 days following the initial nonconformance report. The Caterpillar quality representative (SDE, SQE, buyer etc) will validate the actions taken once implemented.

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CORRECTIVE ACTION (CONTINUED)

Any delays in meeting the timeliness requirements for ECA, ICA and PCA should be discussed with the receiving facility quality representative and the SDE. ICA and PCA completion may extend beyond the timelines where significant change is required, however in all circumstances the timeline should be agreed and understood whilst ensuring that mitigation plans for ongoing preventative inspection continue to support defect-free supply.

The standard process for requesting, communicating, and documenting supplier nonconformances and corrective actions for suppliers to Caterpillar is the EtQ NCM/CAR modules. If this system is not implemented by the purchasing facility, details of the required system will be provided locally, however the basis for any Supplier Corrective action will be based on an 8D format.

When considering corrective action plans you should also consider any systemic causes that could impact other Caterpillar product where similar proactive corrective actions could prevent further defects on other products. As with PCN requests, once corrective actions have been implemented any process documentation that has been updated will need to be submitted for approval by Caterpillar in the form of an updated PPAP.

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DEVIATIONS

Our Quality objective is to receive zero defects from our suppliers; however, it may be required to request deviations from Engineering Drawings, Specifications, or previously approved standards due to identified nonconformances. Deviations are only granted in exceptional circumstances and only by the design owner at Caterpillar.

Deviations are either proactive or reactive with proactive deviations being requested due to a discrepancy being found on the supplier premises prior to shipping material, and reactive deviations being raised due to defective material being found by Caterpillar; whereby suppliers may request deviation to accept material rather than return via transport

Any submitted deviation will be considered, although it may be requested to provide evidence of any mitigating actions taken to minimise the quantity of discrepant material being supplied. Caterpillar uses the deviation module with EtQ reliance unless otherwise directed by the Caterpillar facility requested to accept deviated material. Deviated material should also be clearly identified to avoid secondary inspection once supplied.

COST OF POOR QUALITY (COPQ)

The COPQ program is designed to recover costs incurred by Caterpillar because of supplier delivered nonconformances. The process is defined in MQ4007 and is intended to drive consistency globally when comparing supplier performance and impact to Caterpillar. COPQ recovery is not intended to be punitive or for profit, but to recover the hourly cost value incurred by Caterpillar due to disruption caused by poor Supplier Quality. We would actively encourage a collaborative approach when determining recoveries through COPQ and regular communication on nonconformances, containment and corrective action with Caterpillar is advised.

**For nonconformances found at Caterpillars external customers and dealers COPQ is not applicable, however we would look to recover any Caterpillar costs incurred through warranty.*

DEVIATION PROCESS
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COST OF POOR QUALITY (COPQ)
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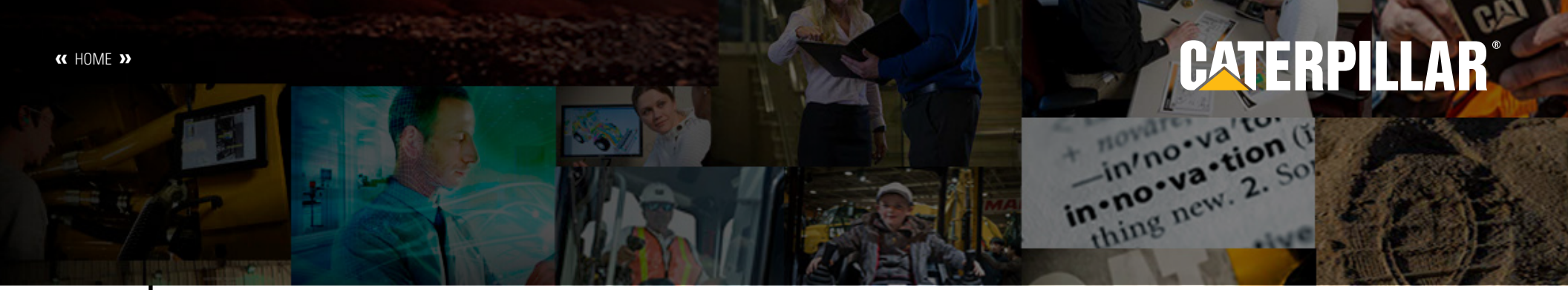

WARRANTY

A warranty process exists to improve quality and the customer experience of our products. When our customers experience issues with our product, our focus is satisfying the customer that root causes have been identified and corrective actions implemented. Rapid identification of solutions to quality and reliability at our customers and dealers is imperative and we expect the same response from suppliers. Warranty recoveries will be made where supplier-caused defects are found at our customers.

CONTINUOUS IMPROVEMENT

Caterpillar has developed a comprehensive lean model focussed on driving continuous improvement and elimination of waste. We have the same expectation that suppliers will embed continuous improvement methodologies through Lean, 6 Sigma or recognised alternative to deliver process and product improvements.

Identifying opportunities to eliminate waste has a mutual benefit for suppliers and Caterpillar. Through ongoing auditing and collaboration, you will have the opportunity to develop processes used within your business through coaching and learning cascaded from Caterpillar. At various stages of this manual, we have introduced auditing and quality processes, which in isolation will indicate improvement actions, but when used collaboratively provide a foundation for the defect-free mindset we require from suppliers.

ROLL OVER TO MAGNIFY IMAGE

WARRANTY MANAGEMENT
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SUPPLIER PERFORMANCE

Managing supplier performance is important to Caterpillar. It is the output of everything we do as a partnership and enables opportunities for development and growth, but also identifies where further actions may be required to manage poor performance. We expect that suppliers will support requested review sessions with the required resources and ensure action plans are executed to agreed deadlines. Where regular meetings are required, agenda items will be agreed prior to each session. A documented pack will be submitted to the Caterpillar representative prior to the session for review. Supplier metrics are available through Supplier Connect and the Assurance of Supply content page.

METRICS

Caterpillar has developed a Supplier Quality Performance Index (SQPI) to capture the performance of suppliers to its globally located facilities. The index takes traditional Parts Per Million (PPM) as a contributor with the addition of rejection frequency, facility rework and down time to capture the overall impact of nonconforming material being supplied. Our quality aim is "zero defects" and when linked to our passion for Lean, the management of supplier performance based on facility impact is fully supported by this SQPI. The SQPI has been developed as our primary leading indicator of Supplier Quality.

Where applicable individual commodities may have metrics aligned to bulk material processing.

Other metrics used for supplier performance management may include:

- PPM
- PPAP on time
- Shipping on time – Production and parts

CORPORATE REJECTION RULES

Whilst other metrics exist for supplier quality monitoring, Caterpillar also acknowledges that Parts per Million (PPM) is a globally recognised indicator of performance and can be used to compare suppliers across the enterprise and for benchmarking new suppliers. It is therefore critical to ensure that pieces rejected due to nonconformances is managed consistently. Caterpillar has developed **MQ4006** as a Manufacturing Practice to be used by receiving facilities to ensure consistency in our approach.

All queries related to facility material rejections should in the first instance be submitted to the Facility Supplier Quality Engineer.

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SUPPLIER PERFORMANCE MANAGEMENT (SPM)

Supplier Performance Management (SPM) is the formal process adopted by Caterpillar and is owned by the purchasing team responsible the supplier. SPM covers five key areas of performance Quality, Cost, Logistics, Development and Management (QCLDM) and has been developed to ensure we manage supplier relationships to meet our goals consistently and on time. Access to the metrics associated to QCLDM will be available to you for analysis and improvement action development. Depending upon your performance, criticality of supply, or exposure to Caterpillar review of your QCLDM metrics will either be monitored, requested, or required to be presented via face-to-face meetings.

SUPPLIER EXCELLENCE RECOGNITION

Throughout this manual we have set our expectations for Supplier Quality. We have detailed how we will engage with suppliers throughout the entire supply life cycle and provided information and links to the tools to achieve excellence.

Supplier Quality is a critical element in our strategy and to encourage and reward excellence we have a recognition program in place to allow suppliers to be certified for the support in our journey. Recognition levels are set at Certified and Excellence, with the potential for suppliers attaining Excellence ratings being nominated for annual Best in Class Awards.

SUPPLIER PERFORMANCE MGMT.
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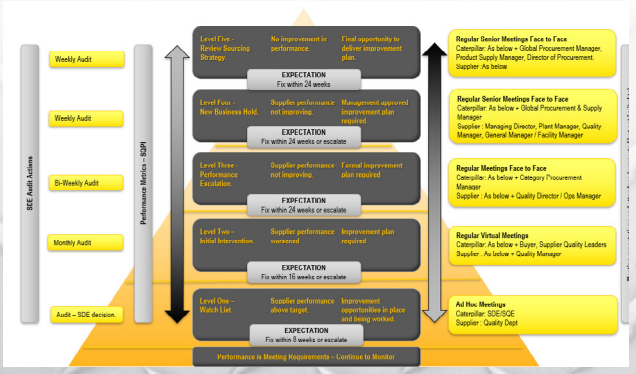
QUALITY ESCALATION PROCESS

Supplier performance is critical to Caterpillar. For Caterpillar to be successful we require excellence from suppliers, and we will monitor performance against set goals to ensure excellence or taking action to improve. Performance against quality goals and the expectations set throughout this manual on occasion may lead to escalation within Caterpillar for further focussed action planning.

If requested, suppliers will be required to have resources available to support action plan development, execution, and communication to Caterpillar. The format and audience for these meetings will be determined based on your performance metrics and status of progress against agreed timelines. If quality performance does not improve, suppliers may be subject to new business hold or loss of business should a systemic failure to demonstrate improvement be found. An example escalation process is set below:



ROLL OVER FOR SUPPLIER ESCALATION MODEL

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GLOSSARY OF TERMS

Below is a list of commonly used terms and acronyms used by Caterpillar either in this guide or by our people.

Advance Product Quality Planning (APQP) – A structured approach to ensure excellence in product quality, through consistent planning, reviewing, and documenting of project steps. Caterpillar uses the AIAG APQP manual as a guide.

Automotive Industry Action Group (AIAG) – Formed in 1982 by the three largest North American Automotive OEMs the AIAG now has over 4000 member companies. This not-for-profit organisation shares procedures and processes to drive cost and complexity from supply chains.

Built in Quality (BIQ) – First stage of Caterpillar Lean deployment and process of managing the flow of defects to ensure quality is built in.

Category – A group of purchased products or services that display similar characteristics and may include design functionality, production process, material content/complexity, supply sources, e.g. (Tyres, Batteries, Castings, Forgings etc).

Code of Conduct – The Code of Conduct reflects the integrity by which a company carries out its business affairs. This integrity encompasses.

Containment – A temporary intervention used to minimise a problems impact on the customer whilst the improvement team look to identify root cause and problem resolution.

Cost of Poor Quality – The costs associated with nonconforming material and poor quality.

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Customer Acceptance Validation (CAV) – An effective critical process control program vital to ensuring defect reduced/free products that aid in creating an acceptable product that will transcend and carry the Caterpillar name to the highest standards in the market.

Defect – An unwanted character present on a product or process; anything that does not meet customer expectations.

Engineering & Procurement Information Exchange (EPIX) – Application for Caterpillar suppliers to view engineering information needed to fulfil purchasing requests.

Failure Mode and Effects Analysis (FMEA) – A formal and structured approach to assess either process or product risk.

Layered Process Audit (LPA) – A Management led process audit following a set agenda to evidence the conformance to documented processes and evidence Continuous Improvement opportunities.

Metrics – A way to provide clear information focussed on key objectives and performance criteria.

New Product Introduction (NPI) – The process Caterpillar uses to develop, build, and deliver quality products and services for customers worldwide. We involve employees, suppliers, dealers, and customers throughout the process.

Parts Per Million (PPM) – Performance metric using material rejection data and a factor or 1 million to give comparable performance irrelevant of supplied volume.

Part Submission Warrant (PSW) – An industry standard document required for all newly tooled or revised products in which a supplier confirms that inspections and tests show conformance with customer requirements. One of the deliverables of the Production Part Approval Process (PPAP).

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Pre-Delivery Inspection (PDI) – An inspection point at the end of the production process intended to validate the customer experience. The intent of PDI is to protect the customer from defects.

Process Flow Diagram – Depicts the flow of material through a process, including any repair or rework processes. A component of the PPAP process.

Production Part Approval Process (PPAP) – Defines requirements for production part approval, including production and bulk materials. The purpose of PPAP is to demonstrate that all customer engineering design record and specification requirements have been understood by the supplier and that the process has the potential to produce product consistently to these requirements during actual production at the quoted run rates. Caterpillar uses the framework for PPAP as defined in the AIAG PPAP Manual.

Quality, Cost, Logistics, Development, Management (QCLDM) – The five components of the relationship measured through the Supplier Relationship Management Process:

- Quality – of products, services, and support.
- Cost – Market competitive, delivering cost reductions.
- Logistics – Deliver what is required, at the right time, in the right place, in the right way.
- Development – Ability to add value and reduce cost and time of development.
- Management – Of a supplier's business and the alignment with Caterpillar expectations, values, strategy, and management.

Suppliers – Providers of, and the party that provides or furnishes an item or service to a purchasing organisation.

1. Production Material
2. Indirect Material
3. Production or Service Parts
4. Services such as heat treatment, plating, polishing, painting etc.

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GLOSSARY OF TERMS

Supplier Excellence Recognition – Caterpillar recognition process whereby having met documented criteria, a supplier may be categorised as either Certified or Excellence.

Supplier Performance Index (SPI) – A measure of overall supplier quality performance using factors that our Global Facilities experience

Supplier Performance Management (SPM) – The framework for managing and developing suppliers in a consistent manner. This process measures supplier performance through Quality, Cost, Logistics, Development and Management.

Supplier Tooling Management System (STMS) – Caterpillar tooling management and requisition system.

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REFERENCES – DOING BUSINESS WITH US

- Quality Policy
- Advanced Product Quality Planning
- 6 Sigma at Caterpillar
- Certification Expectations
- Continuous Improvement
- Contamination Control
- Corrosion Prevention
- Deviation Process
- New Product Introduction
- Nonconformance Management
- Pre Delivery Inspection
- Process Change Notification
- Process Control
- Production Part Approval Process
- Supplier Readiness
- Warranty Management

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